

Ranked Retrieval

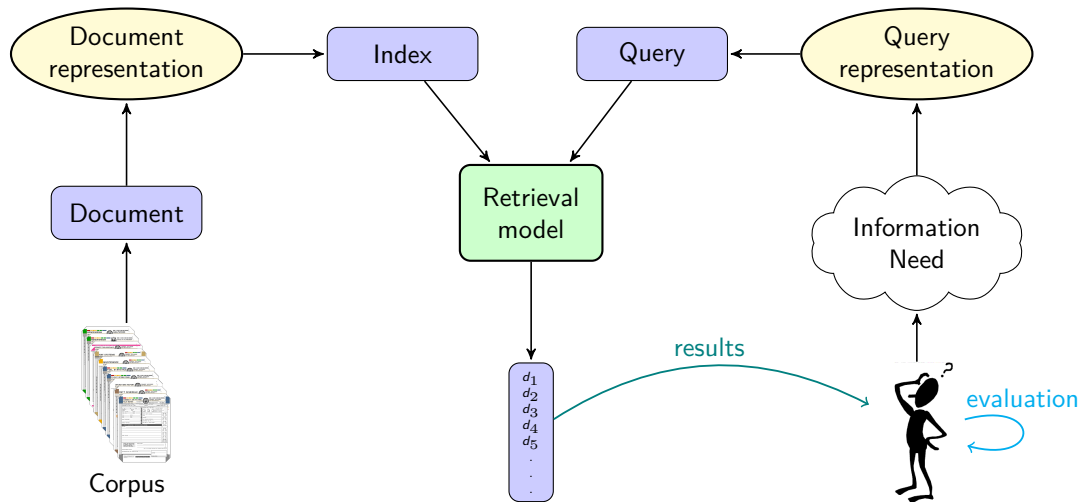
Introduction to Information Retrieval

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Information Retrieval: A Graphical Representation



The Retrieval Task

Input

- A corpus / collection C (set of documents : $d_1, d_2, \dots d_m$)
- A query $Q = \{q_1, q_2, \dots\}$

Output

- A set of documents relevant to the query Q .

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- Boolean retrieval.
- Ranked retrieval.

So far...

Boolean Retrieval

- Make a term-document incidence matrix / inverted index with the documents in the collection.
- User describes their information need using boolean constraints (e.g., **AND**, **OR**, and **AND NOT**).
- Documents either: **match** or **no match**.

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Boolean Retrieval

- Make a term-document incidence matrix / inverted index with the documents in the collection.
- User describes their information need using boolean constraints (e.g., **AND**, **OR**, and **AND NOT**).
- Documents either: **match** or **no match**.
- **Good for expert users** with precise understanding of their needs and the collection.
- **Not good for the majority of users.**
 - ▶ Hard to write boolean queries.
 - ▶ Users don't want to go through 1000s of results.

The Core Problem: From Matching to Ranking

Why Boolean retrieval isn't enough for most users

- **Boolean Retrieval** is a binary decision: MATCH or NO MATCH.
- It fails to distinguish between documents that **all** match.

A Realistic Scenario

Query: climate change impact agriculture

- **Doc A (Focused Report):** Title: Projected Impact of Climate Change on Global Agriculture.
20-page report from the UN where these four terms appear frequently and centrally throughout the text.
- **Doc B (Passing Mention):** Title: Annual Report of the National Science Foundation.
200-page document; on page 150: We also funded a study on broader impact of climate patterns, beyond agriculture; on page 180: This represents a significant change in policy.

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Query: **climate**

- **Doc A** (F) 20-page re text.
- **Doc B** (F) 200-page document;

The Critical Flaw

Both documents are a **Boolean MATCH**. They will be returned in the same result set. A user has to manually sift through potentially thousands of such matches to find **Doc A**.

on page 150: **We also funded a study on broader impact of climate patterns, beyond agriculture**; on page 180: **This represents a significant change in policy.**

Agriculture.
throughout the
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We need a way to **score** and **rank Doc A above Doc B**.

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- **Ranked boolean retrieval**:
 - ▶ retrieve documents satisfying the constraints;
 - ▶ order them based on the number of ways they satisfy the constraints;
 - ▶ not a ranking by **relevance**, but a ranking by **majority of constraints matching**.

Boolean Retrieval

advantage and disadvantage

- **Easy**: from the system's point of view.
- **Transparency**: easy to understand why a document is/isn't selected.
- **Control**: easy to determine whether query too specific (few results) or too broad (many results).

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- **Easy**: from the system's point of view.
- **Transparency**: easy to understand why a document is/isn't selected.
- **Control**: easy to determine whether query too specific (few results) or too broad (many results).
- The burden is on the user to formulate an effective query.
- Need to rank the documents by **relevance**.

Notion of Relevance

- Many factors affect whether a document satisfies a particular user's information need.
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- Predict topical relevance → our focus.

Best-match Retrieval Models

beyond exact-match retrieval models

- Relevance of a document d for a query Q - $\text{rel}(d,Q)$.
- **Best-match** model $\rightarrow$ predicts this relevance based on $\text{similarity}(d,Q)$.

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How to compute $\text{score}(d, q)$?

Ranked Retrieval

using Jaccard Coefficient

- A commonly used metric to measure overlap of two sets A and B :

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 - ▶ $|D_1 \cap D_2|$ -
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- D_1 - John plays football. He also sings with Guitar.
- D_2 - Kate sings, and plays Violin, and Guitar.
 - ▶ $|D_1 \cap D_2|$ -
 - ▶ $|D_1 \cup D_2|$ -
 - ▶ $jaccard(D_1, D_2)$ -

Retrieval with Jaccard Coefficient

example

Example: Query about Making Apple Pie

- **Query Q:** apple pie recipe
 - **D1:**
 - ▶ Content: apple pie recipe sugar flour
 - **D2:**
 - ▶ Content: apple juice recipe
- Calculate the Jaccard coefficient for both. According to Jaccard, which document is more relevant? Does this match your intuition? Why or why not?

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- 👉 Jaccard ranks D2 higher than D1.

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 - 👉 Jaccard ranks D2 higher than D1.
 - 👉 Jaccard penalizing longer documents.

Retrieval with Jaccard Coefficient

first step towards ranking

Example: Query about Python Programming

- **Query Q:** python list comprehension
- **D1:** A short tutorial titled Python List Comprehensions.
 - ▶ Content: python, list, comprehension, syntax, example
- **D2:** A long book chapter Introduction to Python.
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Retrieval with Jaccard Coefficient

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Retrieval with Jaccard Coefficient

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👉 However, it still ignores **term frequency** and **term importance**.

Ranked Retrieval

using Jaccard Coefficient for ranked retrieval

- All document in the collection - considered as bag of words.
- Given query - considered as bag of words.

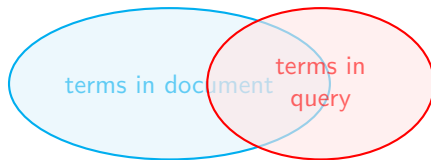


Figure: The bag of words approach to information retrieval.

- For each d in C :
 - ▶ $\text{score}(d, Q) = \text{jaccard}(d, Q)$

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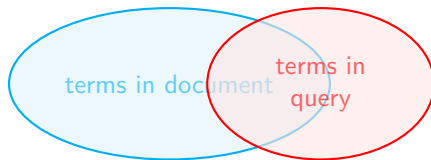


Figure: The bag of words approach to information retrieval.

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Is there any difference between Boolean retrieval and the Jaccard based document scoring?

Ranked Retrieval

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- Treats all terms as equally important.
 - ▶ no notion of term importance.
 - ★ rare terms → more important.
 - ★ frequent terms → less important.
 - ▶ IISER admission procedure → $\text{importance}(\text{admission}) > \text{importance}(\text{procedure})$

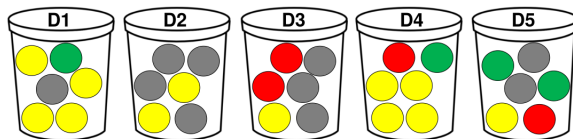
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 - ▶ IISER admission procedure → importance(admission) > importance(procedure)
- No length normalization factor:
 - ▶ $Q = \{q_1, q_2, q_3\}$
 - ▶ $|D1| = 3, |D2| = 1000 \rightarrow$ favours shorter documents.

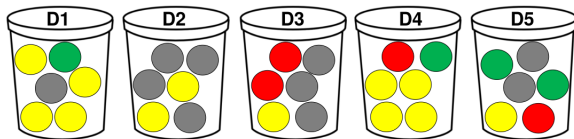
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- Assume - Collection → five documents -
 - ▶ different colored ball → different terms.



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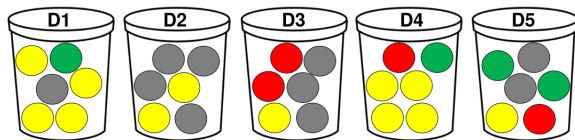
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 - ▶ Which is the most relevant document?
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We need a factor for **term importance**.

Term Importance

next in the agenda

- ✓ Introduction to ranked retrieval.
- ✓ Jaccard similarity and its limitation.

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- ✓ Introduction to ranked retrieval.
- ✓ Jaccard similarity and its limitation.
- Define the notion of term importance.
- TF-IDF term weighting.
 - Vector Space Model.
 - BM25.